Citizen (Community Plant Watering System)

- what should be visual (UI/UX) to the user?
- How to confirm the user did it correctly? trust and sensor data once uploaded via MCU LoRaWAN.
- How to ensure they get the benefits after completion?
- How to do to the tree?

Requirements

- A dashboard, web app, or interactive map that visualizes the hydration status of urban trees (e.g., Green = Good, Yellow = Needs Water Soon, Red = Critical).
- A public-facing element that directs **citizens** to "Red" trees near them.
- low-cost, scalable, and community-driven solution.
  - As community-driven, the municipality will save money on water truck logistics, protecting the urban canopy, reducing CO2, and managing public relations.
  - Wanting an easy, rewarding way to help their neighborhood trees beat the summer heatwaves without wasting their own time or water.
- A mobile or web app with a notification system, geolocation (to find nearby volunteers), and a gamified community board (leaderboards, points for watering plants).
- The trees belong to the city. The city maintains the official *Baumkataster* (the public tree registry database containing the geo-coordinates of over 88,000 urban trees in Karlsruhe). You need the city's permission and data to mount hardware nodes onto public property.

Technical

- **The "Adopt-a-Tree" Model:** Citizens can virtually adopt a specific tree on their street through the web app and name it (e.g., "The Oak of Kriegsstraße"). When the sensor indicates the tree is thirsty, the app sends a friendly push notification: *"Your tree needs 20 liters of water today to stay healthy!"*
- **Immediate Visual Feedback:** The greatest reward for a volunteer is seeing their impact. When a citizen pours a bucket of water over the roots, the moisture level on the public map jumps from 15% (Red/Critical) to 80% (Green/Healthy) in near-real time. They instantly see that *they* saved that tree today.
- **Leaderboards & Badges**: Neighborhoods can compete against each other (e.g., Südstadt vs. Oststadt) to see who keeps their urban canopy the healthiest. Citizens earn badges like "Drought Warrior" or "Canopy Protector" that they can share on social media.
- **The "Green Streak":** Keeping a tree healthy for a consecutive month earns them local recognition in the app or on the city's public smart displays.
- **KVV Public Transit Discounts:** Every 100 liters of documented watering (verified by the sensor's moisture spike immediately following a citizen's tap on the "I am watering this tree" button) earns points redeemable for a discount on a local KVV tram/bus ticket or a free day-pass.

Questions

Cloud (the infastruture)

- What kind of data (from MCU) to use and utilise?
- What kind of (external) data to use? How to implement them?
    - Map / satellite image
    - Weather forecast
    - (what else?)
- How to process the MCU and external data together to build the estimation/predication of when and how much the plant need watering.
    - How to utilise AI for that? do we need AI?
- Where to register these data (Database) and for how long?

Government

- How to implement (UI/UX) the Admin/Main Monitor Dashboard? Previlage?
- (still not fully visioned yet)
- integrate it to smart IoT devices - other devices to open water valve once needed to water the plant automatically.
- Image Processing: Algorithms, computer vision, and heavy data engineering
- Use starlight data to identify the plant need water? (known limitation: it doesn't work wall in dense area)

Requirements:

it **saves money, reduces risks, or optimises operations**.

- We drastically cut municipal irrigation costs and optimize water truck routing through automated data, while boosting public civic engagement.
- a backend tool for **municipalities** to optimise their irrigation truck routes.
- Solve the cities financially unsustainable problem with sending trucks out blindly to water plant(s).
- Create a mock-up map of Karlsruhe using real open data (cities often publish public tree registers as Open Data).
- a robust database for matching profiles
- building clean UI/UX
- extracting insights, automated labelling, or computer vision processing from a dataset of images
- optimised watering routes for the city worker to know the route to plant and amount of water to do in frequently bases. Bases optimal way to save money instead of blind truck watering.
    - Having a dashboard that tells them exactly where to drive their water trucks tomorrow morning based on data, instead of following a fixed, inefficient calendar schedule.

Hardware IoT

Questions

Sensor

- What (type of) sensors do we need to use?
- Where to install the sensor(s)?
- How to connect/install the sensor(s)?
- What data do we need to measure?
- How to ensure plant get water? (reading data)
  - use multiple sensor around the plant? tell user to pure water on a specific area (Where the sensor is located)
- How often do we need to get data from sensor(s)
- what happened if "sensor failure"?

MCU

- What kind of power source to use? battery, solar/wind-source, etc.
- Where to install/mount the MCU? (in the ground, on the tree, etc?)
- How to configure/identify the tree? such as location, and type of plant (to identify the frequents to water it).
- What kind of (Wireless) communication do we need to use? (WIFI, BLE, LoRaWAN, etc.)
- Which area to water?
- How to identify the health of the try?
- How to know it get watered? Like is there a way to read data from the steam of the plant?
- can we get energy from soil? any cutting-edge way?
- How many MCU per plant? can it be possible multiple plants get same MCU?

Hardware integration, IoT:

- **The Enclosure:** Use an **IP65 or IP67-rated weatherproof junction box**. Use **PG7 cable glands** with rubber gaskets to pass the sensor wire out of the enclosure. Tighten them completely to form a liquid-tight seal.
- [Capacitive Soil Moisture Sensor V2.0]()
  - unlike resistive ones, they don't corrode in soil over time.
  - The Capacitive Soil Moisture Sensor V2.0 is a useful tool for gardeners and plant enthusiasts. It makes keeping an eye on soil moisture levels easier, and by making sure your plants get the right amount of water, it helps you take better care of them. You can keep your vegetation

    healthy and flourishing by incorporating this sensor into your routine for caring for plants.
  - **Average Lifespan: 2 to 5 years**, depending on environmental conditions.
  - **Instantaneous response:** The electrical measurement itself is nearly immediate. The onboard timer chip updates the analog signal continuously, meaning the microcontroller can read a new value in a matter of milliseconds. **Real-world response:** While the hardware reads data instantly, the *soil* takes time to shift. When you water a plant, you will see the numbers adjust within **5 to 10 seconds** as the water physically seeps down into the soil surrounding the probe.
- Microcontroller (like an ESP32 or Arduino) with Wi-Fi/Bluetooth capabilities to send the signal.
  - must utilize **ESP32 Deep Sleep**
  - **Ultra-Low-Power (ULP) Co-processor & Deep Sleep:** The ESP32 can shut down its main dual-core processor entirely, dropping current consumption to around **10–15 µA**. It can wake up using an internal timer, read the sensor, transmit, and sleep again.
  - **Built-in Wi-Fi & Bluetooth (BLE):** Ideal for prototyping. BLE allows citizens to interact with the tree node directly via their smartphones when standing next to it.
  - **Analog-to-Digital Converter (ADC):** Reads the continuous voltage output from the soil moisture sensor.
- Connectivity & Data Flow
  - **LoRaWAN (Scalable Production Pitch):** For a true smart city deployment, mention that while the prototype uses Wi-Fi, the production model would use an **ESP32 with LoRaWAN** (connecting to The Things Network). This allows kilometers of range with minimal power.
- Backend & Visualization
  - **Broker/Ingress:** MQTT (e.g., HiveMQ or Mosquitto) or a lightweight HTTP REST API to ingest data.
  - **Database:** A time-series database (like InfluxDB) or a simple Firebase instance to keep track of moisture levels over time.
  - Collected Data: sensor status, battery levels, and ...

Technical:

- The Hardware/electronics need to be insulated and completely waterproof. (from rain and watering to avoid short-circute).
- Calibrate the sensor by recording the analog value in completely dry air versus a glass of water. Implement the Deep Sleep routine.
- Have it send the calibrated moisture percentage to a backend or a free IoT platform like Blynk, ThingSpeak, or Adafruit IO for rapid prototyping.
- In the UI/UX, Overlay your live sensor data onto one of these trees.
- (Capacitive Soil Moisture Sensor V2.0) Proper physical placement ensures accurate readings and protects your hardware:
  - **Depth:** Insert the blade vertically into the soil. Crucially, **do not submerge it past the "Warning Line"** printed on the PCB. The electronic components at the top must stay entirely above the soil line.
  - **Proximity to the Plant:** Place the sensor halfway between the base of the plant stem and the edge of the pot. Avoid shoving it directly into the primary root ball (which can damage the roots) or right against the edge of a plastic pot (which can distort the electrical field).
    - Placing the entire sensor underground is **not recommended** out of the box and will likely cause immediate or rapid failure

    - Set area to water (not anywhere) because if user water the opposite side of the pot, **the sensor will not register the moisture immediately.** Capacitive sensors have a very localized "sensing zone"—they only measure the dielectric constant of the soil within a few centimeters of the blade's surface.
- installed in a public space
    - theft is a valid concern: **Financial Risk:** the full product must be in budget - incredibly cheap so the replacement cost is minimal.
    - fake rocks.
    - heavy object beneath so it cannot be easily pulled out with a simple tug.
- Additional sensor:
    - Soil Temperature (e.g., DS18B20 waterproof probe) - The actual temperature of the root zone. Soil temperature dictates root health and nutrient uptake. It also helps calibrate the moisture readings, as capacitance can shift slightly with temperature.
    - NFC?
    - Add a GPS Module (Hardware-based, High Accuracy). If the plant is out in a remote field, community allotment, or a large farm where there is no Wi-Fi, you can wire a small, cheap GPS module (like the NEO-6M) to the ESP32 via serial communication (UART). **Accuracy:** ~2.5 meters (Pinpoint accuracy anywhere on Earth). **The Catch:** GPS modules take a lot of power and need a clear view of the sky to lock onto satellites. To save battery, you should only power on the GPS module once a day, or *only* when the "Tamper/Theft" vibration switch is triggered.
- Edge-cases: (ESP32 will run a check System Health)
    - Physical Theft or Tamper Detection
    - Battery Issues
        - send a "Low Battery" alert.
    - Cable / Wiring Damage
    - Sensor Faults (Water Damage or Corrosion)
- City truck water the plant from distance. ?
- Wood struture under the ground. Orgamic to install and fix in the soil.